AS1202 – (Group) Coursework:

Index-Linked Bond Cash Flow Model with Interactive UserForm VBA/Excel Program

Part 1: VBA Core Model

You are working in an investment company as a pricing analyst, and you are required to build a VBA pricing and dividends analysis tool (i.e. model program) for index-linked bond assets purchased between ***January 2001*** and ***December 2024***. Consider that you are introducing a new approach to analysing your client's reports by using VBA programming. You want to persuade them to appreciate the advantages of VBA modelling.

1.1 The input parameters to the CM are given as follows:

i. Date of purchase (month and year) and length of remaining term until maturity in whole years (fixed between 10 – 35 years). Optionally (for bonus marks), you can also consider:
   - Optional redemption term (minimum 5 years interval between redemption dates)
   - The date of first coupon payment, which would be treated as ex-dividend if up to 3 months after purchase and with-dividend otherwise

- Nominal amount purchased (between £100,000 - £300,000)
- Purchase price (as a percentage of the nominal amount, between 80% - 120%)
- Income and capital gains tax (CGT) rates (between 20% - 35%)
  - Note: CGT would be applicable only when the purchase price is less than the indexed redemption amount (at maturity)
- Annual coupon rate (as a percentage of the nominal amount, between 5% - 10%)
- Frequency of payments (monthly, quarterly, half-yearly or yearly) – to be entered as text or a recognisable fragment of text (e.g. "mon", "year", etc.)
- Redemption amount (as a percentage of the nominal amount, between 90% - 110%)
- Lag period for the indexation of coupons and redemption (4, 6 or 8 months) – to be entered as an integer value
- Index table(s) by years and months between Jan 1991 to Dec 2022 (e.g. CPI, as provided)
- Future average annual inflation rate (fixed between 2.5% - 7.5%)
  - Required for all <u>future</u> coupon and redemption amounts payable after Jan 2023 (i.e. for all future dates where no index is available in the current table)

1.2 The CM should output the following: [1]

i. Cash flow table (CFT) [2]
   a. For example: CF Dates, applicable index, base coupon/redemption amounts, indexed and taxed coupon/redemption amounts (including purchase price)

ii. Based on the CFT from *part i*, the CM should also estimate the annual effective money yield of the ILB investment. Namely:
   a. Apply linear interpolation of the NPV[3] expression (based on the given CFT) using 0.25% intervals between 0%-20% per annum
   b. Any value outside of the above interval should be displayed as a value error

Part 2: Interactive Interface (UserForm)

In this part of the group-coursework exercise you are required to design a UserForm (UF) interface to the VBA CM (i.e. as described in Part 1). Thus, the purpose of this task is to design and code a customised dialog box that allows the user to enter the CM parameters and to run the program. [4]

2.1 Overall UF Design and Command Buttons:

The UF needs to be designed such that at the start up (e.g. UserForm_Initialize) it reads in the parameter values given in the relevant input WS cells (e.g. next to the CFT) and then allows the user to make changes within the ILB parameter constraints. The UF should have a minimum of three command buttons (CB), corresponding to 'CFT', 'Done' and 'Cancel' actions, as described below:

- The **'CFT'** action CB should check the parameters of the ILB and, if necessary, alert the user of any potential errors. Then, if all the input parameters are given and are correct, it executes the main ILB macro and outputs the CFT together with the current parameters to a NEW worksheet. Finally, it returns the UF display with an updated yield estimate (see yield label control).
- The **'Done'** action CB should be disabled until at least one complete run of the CFT CB action (i.e. with confirmed parameters and output of CFT in a new worksheet). Thus, following a successful run of the CFT the CB should be enabled. The CB action should display a final (brief) summary information about the last executed ILB (e.g. amount of CGT, current CFT worksheet name, or anything else you find useful) and show some closing/goodbye message to the user and exit the UF program.

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[1] **Note: the output of the program should contain a copy of the current input parameters on the same WS (see section 2.2).**

[2] See on Moodle numerical examples about the calculation of the cash flow components of the balance sheet and model outputs.

[3] Provide your own UDF to calculate the NPV, instead of the built-in VBA NPV function!

[4] Ideally, there should be a purpose-built macro outside of the UF program to check the parameter settings which is called by the main macro (or CFT see section 2.1).

- The **'Cancel'** CB action should request a confirmation from the user that indeed wants to cancel (i.e. exit) the program. Then, in case of a 'Yes' response, it resets the (input) WS parameter values to their last correct settings used for the CFT (i.e. either the start-up values or those used in the last CFT run) and exits the UF. Alternatively, in the case of a 'No' response, it returns to the current (active) UF and waits for further user inputs.

2.2 Technical UF Specifications:

The UF should satisfy the following (*minimum*) technical specifications:

1. Declare in the program module a number of Public (i.e. global) variables corresponding to the main ILB parameters, which will be used as the current values at CFT execution time (i.e. the same as the ones you have probably made use of in the coursework 1 program).
2. In addition, declare in the program module a number of Public (i.e. global) variables corresponding to the main ILB parameters, which will be used as temporary values in-between CFT runs (i.e. a new set of variables that act as copies of the current ILB parameters, which are used at Cancel). Thus, after every successful CFT run these temporary variables will be overwritten (i.e. reset) by the current CFT variables.
3. Include on the UF a (clearly visible) ***label*** control that displays the estimated annual yield. The label should display a temporary message (e.g. Pending, Unknown, Run CFT, etc) until the execution of the CFT CB. Thus, practically, the label control text should show a yield value only immediately after the CFT run. Then, consecutively, it should become unknown following any parameter change on the UF.
4. All the other parameters (i.e. controls) on the UF should be linked to their corresponding cells in the input WS (e.g. placed on the main WS).
5. There should be at least **3** parameters that have multiple controls on the UF that are linked up (e.g. spin button and textbox/label, or spin button and scroll bar, etc).

The UF should be designed such that it is aesthetically appealing (i.e. well-proportioned and has an ordered layout) and it is easily understandable by a user. Additionally, you might consider other good/helpful functionalities, like: tab order, pointer tips, extra error traps, help button, etc..

Part 3: Coursework Tasks

Therefore, you need to put together an Excel ME-WB that will aid the pricing and analysis of an appropriate ILB investment.[5] Thus, you should design a VBA CM with a UF interface that can be ran interactively by a user. The model inputs/outputs should be handled entirely through the application of the UF (i.e. making use of the UF input fields and CBs). The UF

[5] Make use of the template cw file on Moodle: AS1202_T2 - CW_3-GR-.xlsm ! Fill in the two template modules and add a UF module.

should be able to validate the input parameters from section 1.1 [6] (automatically detect or trap input errors both at the start-up of the UF and during the running of the UF[7]). Further, it should execute the required actions through the CBs (see section 2.1) and run the yield estimation UDF.

Finally, you need to provide a brief written description of your overall program structure in the Module 'CW3_GR_Set' in the form of VBA comments. Describe the main features of both the CM and of the UF, and how they have been implemented (e.g. global/private variables, type of arguments that the UDFs/macros might require and what they represent, dependencies, etc.).[8]

Other than that, you might describe here any additional features that you have managed to implement in the program (e.g. other than the above minimum specifications, such as the price of a corresponding standard bond (i.e. unindexed) with the given estimated yield, etc.).

Furthermore, each UDF/macro should be documented with brief comments at their own definition section regarding any special considerations and/or how the code works (e.g. above the declaration statements of the routines). Similarly, there should be sufficient comments at each control event handlers regarding any special considerations and/or how the code works.

Finally, you need to provide a supplementary ***'user guide'*** (i.e. written and illustrated description of your program). This should contain an overall description and instructions how to use the program (e.g. input parameters, run the model and view/collect the results, etc.). Further, there should be illustrations of potential errors (e.g. troubleshooting) and any other functionalities that you might have implemented (i.e. outside of the above minimum specifications). [9]

The assessment will be based on the level of basic comprehension and coding skills that you demonstrate, including your ability to apply the relevant VBA concepts.

The efficiency, elegance and clarity of the code will be credited. Please remember to:

- Use the most efficient commands and structures.
- Choose your variable names and data types carefully.
- Present carefully and indent your code.
- Avoid overusing WorksheetFunction calls to Excel built-in functions (the less you use them the better).
- Comment sufficiently throughout to explain your code.

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[6] Note, however, the CM should only run through the CFT button of the UF if *ALL* the parameters have been successfully validated.

[7] **WARNING**: For the purposes of the VBA project, it IS NOT allowed to make use of Excel data validation methods (e.g. as commonly applied in the AS1104 CW2)! Thus, both the parameter validation and the error traps should be entirely VBA-based within the UF validation routines.

[8] See further hints and instructions therein. You can think of this as a description of the program's flow chart.

[9] **Warning**: you DO NOT need to describe how you have constructed and coded the UF!

User Guide & VBA Documentation	Excel-VBA Model (*.xlsm file)				Bonus
	Core Model		User Form		
	Method & Code structure	Misc CFT & yield estimate	Layout & Main Controls	Auxiliary Controls	
25	20	20	25	10	10

Part 4: Submission

Submit on Moodle a macro enabled Excel WB with your VBA program.

You should name the file as "AS1202_T2 – CW_3-GR-yy.xlsm", where yy is your group number!

There should be only a single submission by each group before the deadline on Moodle. Note that any submission after this deadline will be subject to a 10% penalty per day on the coursework grade!

Don't forget that this is a peer-assessment (PA) type group coursework whereas everyone needs to rate their teammates. The PA should be honest and should reflect on aspects such as: *availability, organisation, co-operation with the rest of the team, creativity, meeting deadlines and competency in their roles* (hence keeping a good track record of the groupwork will be helpful). '*No contribution*' type feedback will be scrutinised and will only be accepted when there are genuine reasons for them.

Please read carefully the documentation on the PA portal before appraising your teammates.